1. This question carries 40% of the mark.

(a) Consider each of the following continuous-time signals. For each case, specify if the signal is causal/non-causal, periodic/non-periodic, odd/even. If the signal is periodic specify its period.

(i) $x(t) = 2\cos(2\pi t)$ **[2]**

Answer

Non-causal, periodic with period $T = 1$, even

Most students answered this question correctly.

(ii) $x(t) = 5e^{2\text{sign}(t)\cdot t}$ **[2]**

Answer

Non-causal, non-periodic, even

A fairly good number of students answered this question correctly. However, a substantial number of students answered wrongly that this signal is neither even not odd.

(b) Consider the signal

$$x(t) = \begin{cases} 1-t, & 0 \le t \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Now sketch each of the following signals and describe briefly in words how each of the signals can be derived from the original signal $x(t)$.

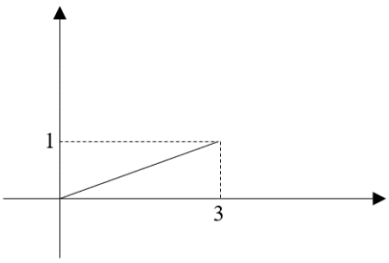
(i) $x\left(-\frac{t}{3}+1\right)$ **[2]**

(ii) $x(2t+5)$ **[2]**

Answer for (i) and (ii)

$$x\left(-\frac{t}{3}+1\right)$$

$$= \begin{cases} 1-\left(-\frac{t}{3}+1\right) = \frac{t}{3}, & 0 \le -\frac{t}{3}+1 \le 1 \Rightarrow -3 \le -t \le 0 \Rightarrow 0 \le t \le 3 \\ 0 & \text{otherwise} \end{cases}$$

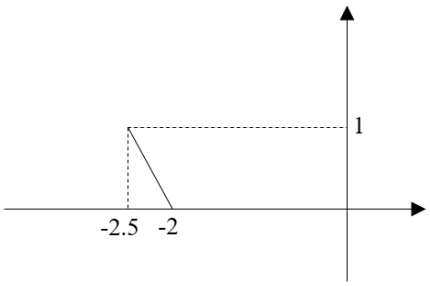


Most students answered this question correctly.

$$x(2t+5)$$

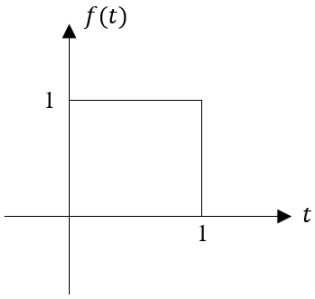
$$= \begin{cases} 1-(2t+5) = -2t-4, & 0 \le 2t+5 \le 1 \Rightarrow -5 \le 2t \le -4 \Rightarrow -\frac{5}{2} \le t \le -2 \\ 0 & \text{otherwise} \end{cases}$$

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Most students answered this question correctly.

(c) By applying the property of convolution, find the system's output (i.e., zero-state response) given that the impulse response is $h(t) = e^{-t}u(t)$ and that the input $f(t)$ is as shown in Figure 1(c) below.



[6]

Figure 1(c)

Answer

$h(t) = e^{-t}u(t)$

$f(t) = \begin{cases} 1, & 0 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$

$y(t) = h(t) * f(t)$

$= \int_{-\infty}^{\infty} h(\tau)f(t-\tau)d\tau \Rightarrow y(t) = \int_{-\infty}^{\infty} e^{-\tau}u(\tau)f(t-\tau)d\tau = \int_{0}^{\infty} e^{-\tau}f(t-\tau)d\tau$

$f(t-\tau) = \begin{cases} 1, & 0 \le t-\tau \le 1 \Rightarrow -1 \le \tau - t \le 0 \Rightarrow t-1 \le \tau \le t \\ 0 & \text{otherwise} \end{cases}$

For $t < 0$ the convolution is zero.

For $t - 1 \le 0 \le t$ which means $0 \le t \le 1$ we have

$y(t) = \int_0^t e^{-\tau}d\tau = 1 - e^{-t}$

For $t \ge 1$ we have $y(t) = \int_{t-1}^{t} e^{-\tau}d\tau = e^{-(t-1)} - e^{-t} = e^{-t}(e-1)$

Therefore, the convolution is

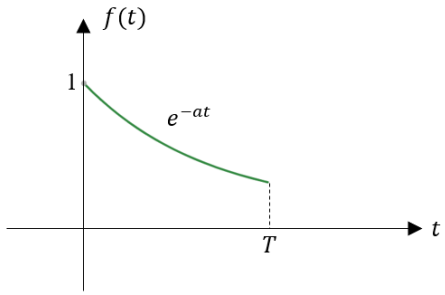
$y(t) = \begin{cases} 1 - e^{-t}, & 0 \le t \le 1 \\ e^{-t}(e-1) & t \ge 1 \end{cases}$



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A fairly good number of students answered this question correctly. However, a substantial number of students provided either an incomplete or a wrong answer.

(d) Derive the Fourier transform of the signals $f(t)$ shown in Figures 1(d)(i)-(ii) below.
(i)



[3]

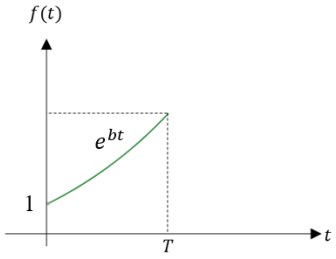
Figure 1(d)(i)

Answer

$$F(\omega) = \int_0^T e^{-at}e^{-j\omega t}dt = \int_0^T e^{-(a+j\omega)t}dt = \frac{1-e^{-(a+j\omega)T}}{a+j\omega}$$

Most students answered this question correctly.

(ii)



[3]

Figure 1(d)(ii)

Answer

$$F(\omega) = \int_0^T e^{bt}e^{-j\omega t}dt = \int_0^T e^{(b-j\omega)t}dt = \frac{1-e^{(b-j\omega)T}}{-b+j\omega}$$

Most students answered this question correctly.

(e) The output $y(t)$ of a continuous-time LTI system is related to the input $x(t)$ by the differential equation

$$\frac{dy(t)}{dt} + 2y(t) = 2x(t)$$

Determine the frequency response of the system and sketch the asymptotic behavior of its Bode plots. **[5]**

Answer

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The transfer function is:

$$H(s) = \frac{2}{s+2}$$

We substitute $s = j\omega$. Then, we obtain $H(j\omega) = \frac{2}{j\omega+2} = \frac{1}{\frac{j\omega}{2}+1}$.

Amplitude response: $|H(j\omega)| = \frac{1}{\sqrt{\frac{\omega^2}{4}+1}}$.

Phase response: $\angle H(j\omega) = \Phi(\omega) = -\tan^{-1}\left(\frac{\omega}{2}\right)$

The log amplitude is $-20\log\left|1+\frac{j\omega}{2}\right|$

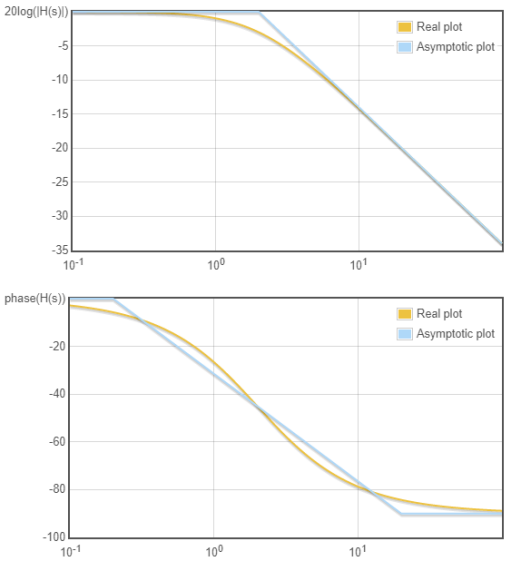
- $\omega \ll 2 \Rightarrow -20\log\left|1+\frac{j\omega}{2}\right| \approx -20\log 1$
- $\omega \gg 2 \Rightarrow -20\log\left|1+\frac{j\omega}{2}\right| \approx -20\log(\frac{\omega}{2}) = -20\log\omega + 20\log 2$

This represents a straight line (when plotted as a function of u, the log of ω) with a slope of $-20dB/$decade or $-20(0.301) = -6.02dB/$octave.
The exact log amplitude for this pole is:

$$-20\log\left|1+\frac{j\omega}{2}\right| = -20\log\left(1+\frac{\omega^2}{4}\right)^{\frac{1}{2}} = -10\log\left(1+\frac{\omega^2}{4}\right)$$

For the phase we have:

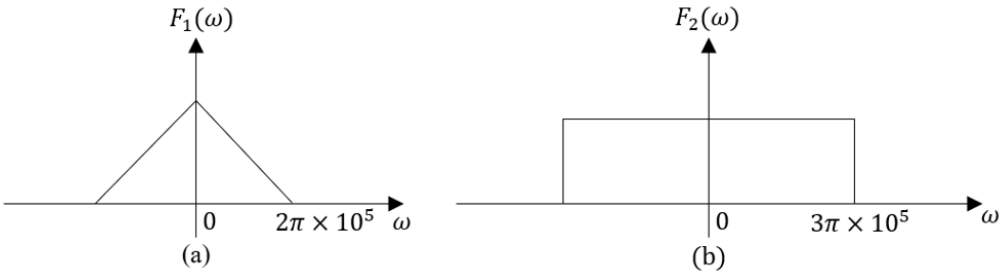
- $\angle H(j\omega) = -\angle\left(1+\frac{j\omega}{2}\right) = -\tan^{-1}\left(\frac{\omega}{2}\right)$.
- $\omega \ll 2 \Rightarrow \frac{\omega}{2} \ll 0 \Rightarrow -\tan^{-1}\left(\frac{\omega}{2}\right) \approx 0$
- $\omega \gg 2 \Rightarrow \frac{\omega}{2} \gg 0 \Rightarrow -\tan^{-1}\left(\frac{\omega}{2}\right) \approx -90^{o}$



About 50% of students produced a fairly accurate pair of plots. The rest of the students produced plots similar to the correct ones but not accurate. Many students provided the Bode plots without any mathematical analysis at all.

(f) Figures (a) and (b) below show the Fourier spectra of signals $f_1(t)$ and $f_2(t)$. Determine the Nyquist sampling rates for signals $f_1(t)$ and $f_2(t)$. (Hint: Use the frequency convolution and the width property of the convolution.)

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(i) $f_1(t)$ **[2]**

Answer

Bandwidth of $f_1(t)$ is 100kHz and Nyquist rate is 200kHz.

(ii) $f_2(t)$ **[2]**

Answer

Bandwidth of $f_2(t)$ is 150kHz and Nyquist rate is 300kHz.

(iii) $f_1^2(t)$ **[2]**

Answer

The Fourier Transform of $f_1^2(t)$ is $\frac{1}{2\pi}F_1(\omega) * F_2(\omega)$. From the property regarding the size of convolution we see that the bandwidth of $f_1^2(t)$ is twice the bandwidth of $f_1(t)$, i.e., 200kHz. The Nyquist rate is therefore, 400kHz.

(iv) $f_2^3(t)$ **[2]**

Answer

Based on (iii) the bandwidth of $f_2^3(t)$ is three times the bandwidth of $f_1(t)$, i.e., 450kHz. The Nyquist rate is therefore, 900kHz.

(v) $f_1(t)f_2(t)$ **[2]**

Answer

Based on the above the bandwidth of $f_1(t)f_2(t)$ is the sum of the individual bandwidths, i.e., 250kHz. The Nyquist rate is therefore, 500kHz.

A fairly good number of students answered this question correctly. A substantial number of students expressed everything using radial frequency; this is wrong.

(g) Consider the discrete-time, causal LTI system with input $x[n]$ and output $y[n]$ related with the difference equation:

$$y[n] - \frac{5}{6}y[n-1] + \frac{1}{6}y[n-2] = x[n] - \frac{1}{2}x[n-1]$$

(i) Find the analytical expression and the Region of Convergence (ROC) of the z −transform of the impulse response of the above system.
[Hint: Use the fact that the z-transform $\frac{z}{z-a}$ corresponds to the function $a^n u[n]$ if $|z| > |a|$ and the function $-a^n u[-n-1]$ if $|z| < |a|$. The function $u[n]$ is the discrete-time unit step function.] **[3]**

(ii) Find the analytical expression and the Region of Convergence (ROC) of the z −transform of the output if $x[n] = \left(\frac{1}{2}\right)^n u[n]$. **[2]**

Answer

(i)

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$$Y(z) - \frac{5}{6}z^{-1}Y(z) + \frac{1}{6}z^{-2}Y(z) = X(z) - \frac{1}{2}z^{-1}X(z)$$

$$Y(z)(1 - \frac{5}{6}z^{-1}Y(z) + \frac{1}{6}z^{-2}) = X(z)(1 - \frac{1}{2}z^{-1})$$

$$Y(z)(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1}) = X(z)(1 - \frac{1}{2}z^{-1}) \Rightarrow$$

$$Y(z)\left(1 - \frac{1}{3}z^{-1}\right) = X(z) \Rightarrow \frac{Y(z)}{X(z)} = \frac{1}{\left(1 - \frac{1}{3}z^{-1}\right)} \Rightarrow H(z) = \frac{z}{z - 1/3}$$

$h[n] = \left(\frac{1}{3}\right)^n u[n]$, $|z| > 1/3$, causal system

$h[n] = -\left(\frac{1}{3}\right)^n u[-n-1]$, $|z| < 1/3$

(ii)

$X(z) = \frac{z}{z-1/2}$, $|z| > 1/2$

$Y(z) = \frac{z^2}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}$, ROC: $|z| > \frac{1}{2} \cap |z| > \frac{1}{3} = |z| > \frac{1}{2}$

A fairly good number of students answered this question correctly. A substantial number of students made mistakes in the calculation of ROC in (ii).

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2. This question carries 30% of the mark.

(a) (i) Find the analytical expression and the Region of Convergence (ROC) of the Laplace transform of the continuous-time, causal signal $x(t) = e^{-at}u(t)$, with a real and positive and $u(t)$ the continuous-time unit step function. **[3]**

Answer

$$X(s) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-st}dt = \int_{0}^{\infty} e^{-at}e^{-st}dt = \frac{e^{-(s+a)t}}{-(s+a)}\Big|_0^{\infty} = \frac{e^{-(s+a)\infty}}{-(s+a)} + \frac{1}{s+a}$$

$$X(s) = \frac{1}{s+a}, \text{ ROC: } \mathrm{Re}\{s\} > -a$$

(ii) Find the analytical expression and the Region of Convergence (ROC) of the Laplace transform of the continuous-time, anti-causal signal $x(t) = -e^{-at}u(-t)$, with a real and positive and $u(t)$ the continuous-time unit step function. **[3]**

Answer

$$X(s) = \int_{-\infty}^{\infty} -e^{-at}u(-t)e^{-st}dt = \int_{-\infty}^{0} -e^{-at}e^{-st}dt = \frac{e^{-(s+a)t}}{(s+a)}\Big|_{-\infty}^{0} = \frac{1}{s+a}$$

$$X(s) = \frac{1}{s+a}, \text{ ROC: } \mathrm{Re}\{s\} < -a$$

(iii) Show that the Laplace transform of the signal $\delta(t)$ is 1 with $\delta(t)$ the continuous impulse function. **[3]**

Answer

$$X(s) = \int_{-\infty}^{\infty} \delta(t)e^{-st}dt = e^{-s\cdot 0} = 1$$

A fairly good number of students answered this question correctly. A substantial number of students provided the Laplace transform of the Dirac function without any intermediate steps.

(b) Consider a continuous-time causal signal $x(t)$, $x(t) = 0$, $t < 0$ with Laplace transform $X(s) = \frac{s+2}{s-2}$. Find the analytical expression of $x(t)$, using the results obtained in question (a) above and also find the region of convergence of $X(s)$.

[Hint: you may write the above Laplace transform as $X(s) = 1 + \frac{4}{s-2}$]. **[4]**

Answer

$X(s) = 1 + \frac{4}{s-2} \Rightarrow$

$x(t) = \delta(t) + 4e^{2t}u(t)$

ROC: $\mathrm{Re}\{s\} > 2$

Some students failed at providing the causal signal and provided the non-causal signal that has the same Laplace transform.

(c) Consider a continuous Linear Time-Invariant system with input the signal $x(t) = \delta(t)$, with Laplace and output the signal $y(t) = -\frac{2}{3}e^{2t}u(-t) + \frac{1}{3}e^{-t}u(t)$.

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(i) Determine the transfer function of the system $H(s)$ and its Region of Convergence (ROC) **[5]**

Answer

$y(t) = \frac{2}{3}e^{2t}u(-t) + \frac{1}{3}e^{-t}u(t)$

$Y(s) = \frac{2}{3}\frac{1}{s-2} + \frac{1}{3}\frac{1}{s+1} = \frac{1}{3}\frac{3s}{(s+1)(s-2)} = \frac{s}{(s+1)(s-2)}$

ROC: $\text{Re}\{s\} < 2 \cap \text{Re}\{s\} > -1 \Rightarrow -1 < \text{Re}\{s\} < 2$

In that case

$H(s) = Y(s)$

A substantial number of students answered this question correctly. Some students failed at providing a correct ROC.

(ii) Determine the Laplace transform of the output of the system and its Region of Convergence (ROC) when the input is the signal $x(t) = u(t)$ (the continuous-time unit step function). The ROC of $Y(s)$ is the intersection of the ROCs of $X(s)$ and $H(s)$. **[5]**

Answer

Consider the signal $x(t) = u(t)$.

$X(s) = \int_{-\infty}^{\infty} u(t)e^{-st}dt = \int_{0}^{\infty} e^{-st}dt = \frac{e^{-st}}{-s}\Big|_0^{\infty} = \frac{e^{-s\infty}}{-s} - \frac{e^{-s0}}{-s} = \frac{1}{s}$, $\text{Re}\{s\} > 0$

$Y(s) = H(s)X(s) = \frac{1}{(s+1)(s-2)}$, $0 < \text{Re}\{s\} < 2$

A substantial number of students answered this question correctly. Some students failed at providing a correct ROC.

(d) The output $y(t)$ of an LTI system is related to the input $x(t)$ by the differential equation

$$\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} - 2y(t) = \frac{dx(t)}{dt} + 2x(t)$$

Let $X(s)$ and $Y(s)$ denote the Laplace transforms of $x(t)$ and $y(t)$, respectively, and let $H(s)$ denote the Laplace transform of the system's impulse response $h(t)$.

(i) Determine $H(s)$ as a ratio of two polynomials. **[3]**

Answer

$(s^2 + s - 2)Y(s) = sX(s) + 2X(s) \Rightarrow (s+2)(s+1)Y(s) = (s+2)X(s) \Longrightarrow$

$\frac{Y(s)}{X(s)} = \frac{1}{s-1} = H(s)$

A substantial number of students answered this question correctly. However, a fairly good number of students failed to reach the final simple form of the transfer function, often due to miscalculations.

(ii) Determine $h(t)$ for each of the following cases:

1. The system is stable.
2. The system is causal.

[4]

Answer

$h(t) = -e^{t}u(-t)$, ROC: $\text{Re}\{s\} > 1$, stable

$h(t) = e^{t}u(t)$, ROC: $\text{Re}\{s\} > 1$, causal

A substantial number of students answered this question correctly.

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3. This question carries 30% of the mark.

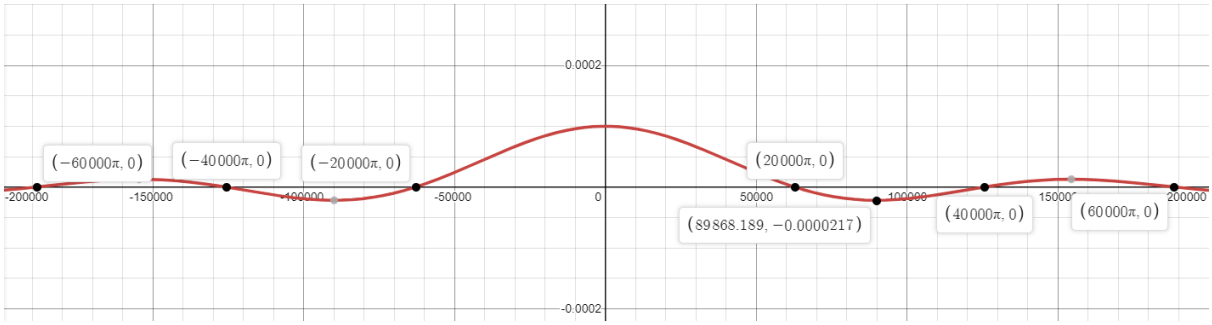
(a) Signals $f_1(t) = \Pi(10^4 t)$ and $f_2(t) = \frac{1}{2}(\delta(t - 0.5 \cdot 10^{-4}) + \delta(t + 0.5 \cdot 10^{-4}))$ are applied at the inputs of ideal lowpass filters with frequency responses $H_1(\omega) = 10^4 \Pi\left(\frac{\omega}{40000\pi}\right)$ and $H_2(\omega) = \Pi\left(\frac{\omega}{20000\pi}\right)$ respectively, with

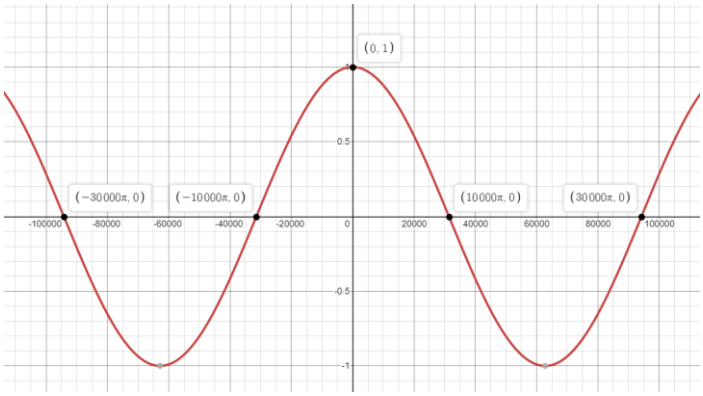
(i) Sketch $F_1(\omega)$ and $F_2(\omega)$. **[2]**

Answer

$$f_1(t) = \Pi(10^4 t) = \begin{cases} 1 & |10^4 t| < 0.5 \Rightarrow -0.5 * 10^{-4} < t < 0.5 * 10^4 \\ 0.5 & |t| = 0.5 * 10^4 \\ 0 & \text{otherwise} \end{cases}$$

The Fourier transform of $f_1(t) = \Pi(10^4 t) = \Pi\left(\frac{t}{10^{-4}}\right)$ is $F_1(\omega) = 10^{-4}\text{sinc}\left(\frac{10^{-4}\omega}{2}\right)$.
The Fourier transform of $f_2(t) = \frac{1}{2}(\delta(t - 0.5 \cdot 10^{-4}) + \delta(t + 0.5 \cdot 10^{-4}))$ is $F_2(\omega) = \frac{1}{2}\left(e^{-j0.5 \cdot 10^{-4}\omega} + e^{j0.5 \cdot 10^{-4}\omega}\right) = \cos(0.5 \cdot 10^{-4}\omega)$.





A substantial number of students answered this question correctly.

(ii) Sketch $H_1(\omega)$ and $H_2(\omega)$. **[2]**

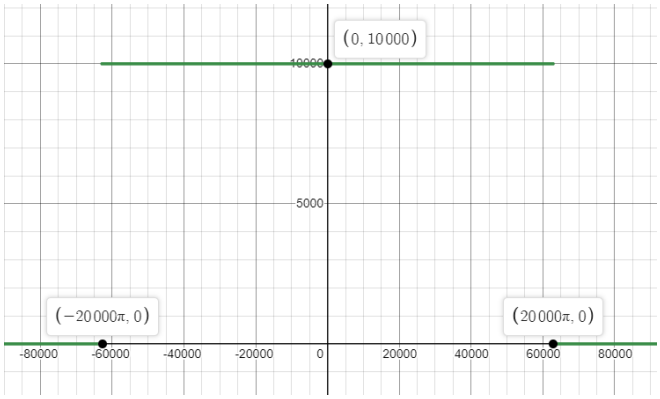
Answer

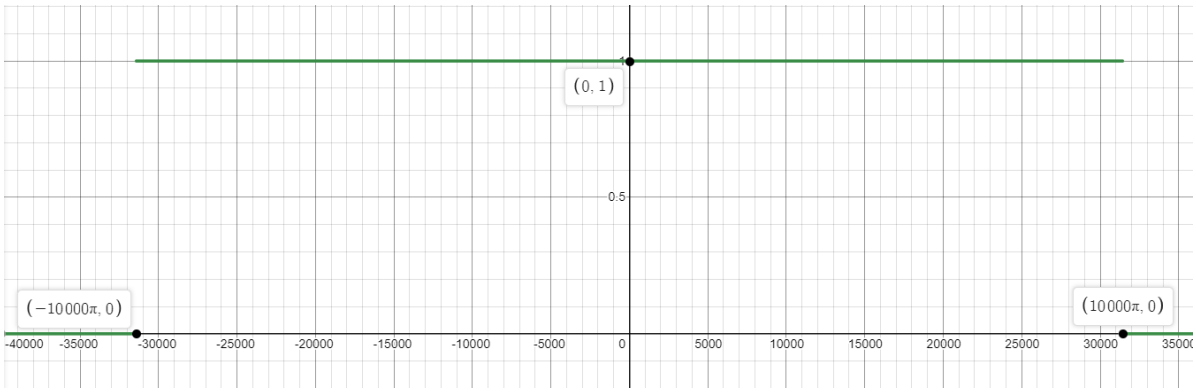
$$H_1(\omega) = 10^4 \Pi\left(\frac{\omega}{40000\pi}\right)$$

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$$= \begin{cases} 10^4 & \left|\frac{\omega}{40000\pi}\right| < 0.5 \Rightarrow -20000\pi < \omega < 20000\pi \\ 0.5 * 10^4 & |\omega| = 20000\pi \\ 0 & \text{otherwise} \end{cases}$$

$$H_2(\omega) = \Pi\left(\frac{\omega}{20000\pi}\right) = \begin{cases} 1 & \left|\frac{\omega}{20000\pi}\right| < 0.5 \Rightarrow -10000\pi < \omega < 10000\pi \\ 0.5 & |\omega| = 10000\pi \\ 0 & \text{otherwise} \end{cases}$$





A substantial number of students answered this question correctly. However, many students put 1 as the amplitude of the rect function in both cases.

(iii) Sketch $Y_1(\omega)$ and $Y_2(\omega)$. **[2]**

Answer

$$Y_1(\omega) = F_1(\omega)H_1(\omega)$$

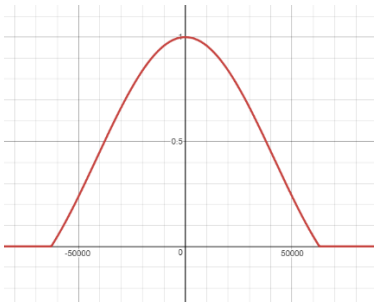
$$= \begin{cases} 10^4 10^{-4}\text{sinc}\left(\frac{10^{-4}\omega}{2}\right) & \left|\frac{\omega}{40000\pi}\right| < 0.5 \Rightarrow -20000\pi < \omega < 20000\pi \\ 0.5 * 10^4 10^{-4}\text{sinc}\left(\frac{10^{-4}\omega}{2}\right) & |\omega| = 20000\pi \\ 0 & \text{otherwise} \end{cases}$$

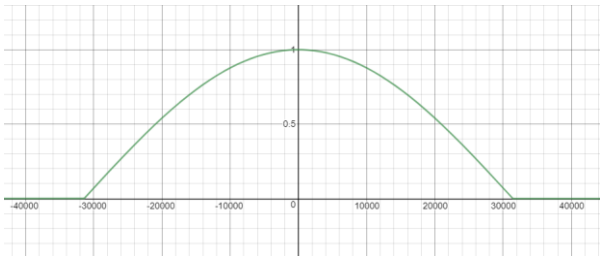
$$= \begin{cases} \text{sinc}\left(\frac{10^{-4}\omega}{2}\right) & \left|\frac{\omega}{40000\pi}\right| < 0.5 \Rightarrow -20000\pi < \omega < 20000\pi \\ 0.5 * \text{sinc}\left(\frac{10^{-4}\omega}{2}\right) & |\omega| = 20000\pi \\ 0 & \text{otherwise} \end{cases}$$

$$Y_2(\omega) = F_2(\omega)H_2(\omega)$$

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$$= \begin{cases} \cos(0.5 \cdot 10^{-4}\omega) & \left|\frac{\omega}{20000\pi}\right| < 0.5 \Rightarrow -10000\pi < \omega < 10000\pi \\ 0.5\cos(0.5 \cdot 10^{-4}\omega) & |\omega| = 10000\pi \\ 0 & \text{otherwise} \end{cases}$$





A substantial number of students provided almost correct plots.

(iv) Find the Nyquist sampling rate of $y_1(t)$, $y_2(t)$ and $y(t)$. **[2]**

Answer
The bandwidth of $y_1(t)$ is 10kHz.
The bandwidth of $y_2(t)$ is 5kHz.
The bandwidth of $y_1(t)y_2(t)$ is 15kHz. The Nyquist rate is 30kHz.

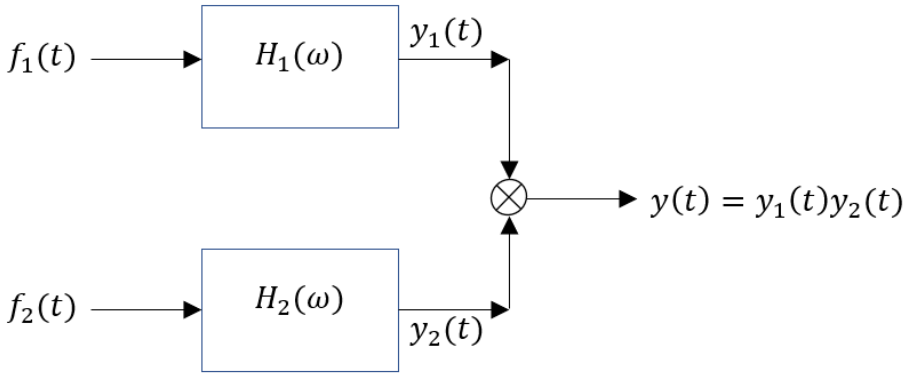


Figure 3a

A substantial number of students provided correct answers. A fairly good number gave their answers using radial frequencies and as previously mentioned, this is not right.

(b) (i) Consider a discrete signal $x[n]$ with z −transform $X(z)$. Find the z −transforms of the signals $x[n+3]$ and $x[-n+1]$ as functions of the z-transform $X(z)$. **[5]**

Answer
Assume that $x_1[n] = x[n+3]$.

$$X_1(z) = \sum_{n=-\infty}^{\infty} x_1[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[n+3]z^{-n} = \sum_{n=-\infty}^{\infty} x[n+3]z^{-(n+3)}\, z^3 = z^3X(z)$$

Assume that $x_2[n] = x[-n+1]$.

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$$X_2(z) = \sum_{n=-\infty}^{\infty} x_2[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[-n+1]z^{-n} = \sum_{n=-\infty}^{\infty} x[-n+1]\left(\frac{1}{z}\right)^{n}\left(\frac{1}{z}\right)^{-1}\frac{1}{z}$$
$$= \frac{1}{z}\sum_{n=-\infty}^{\infty} x[-n+1]\left(\frac{1}{z}\right)^{-(-n+1)} = \frac{1}{z}\sum_{k=\infty}^{-\infty} x[k]\left(\frac{1}{z}\right)^{-k} = \frac{1}{z}X\left(\frac{1}{z}\right)$$

A substantial number of students provided a correct answer for the first function. Unfortunately, very few students provided a correct answer for the second function.

(ii) Consider a signal $y[n]$ which is related to two signals $x_1[n]$ and $x_2[n]$ by

$$y[n] = x_1[n+3] * x_2[-n+1]$$

where

$$x_1[n] = \left(\frac{1}{2}\right)^n u[n] \text{ and } x_2[n] = \left(\frac{1}{3}\right)^n u[n]$$

with $u[n]$ the discrete unit step function. Use the results obtained in question (b)(i) above to determine the z-transform of $y[n]$.

[5]

Answer

$x_1[n+3] \neq 0, n \geq -3$

$$Z\{x_1[n+3]\} = z^3 X_1(z) = z^3 \frac{1}{1-\frac{1}{2}z^{-1}} = z^3 \frac{z}{z-\frac{1}{2}} = z^4 \frac{1}{z-\frac{1}{2}}$$

$$Z\{x_2[-n+1]\} = \frac{1}{z}X_2\left(\frac{1}{z}\right) = \frac{1}{z(1-\frac{1}{3}z)}$$

$$Y(z) = Z\{x_1[n+3]\} \cdot Z\{x_2[-n+1]\} = z^4 \frac{1}{\left(z-\frac{1}{2}\right)} \cdot \frac{1}{z\left(1-\frac{1}{3}z\right)}$$

$$= -6z^3 \frac{1}{(2z-1)} \cdot \frac{1}{(z-3)}$$

A substantial number of students provided a correct approach to this question. However, due to wrong results in the previous question they came up with wrong answers in this question as well.

(c) Consider an even sequence $x[n]$ (i.e., $x[n] = x[-n]$) with rational z-transform $X(z)$.

(i) From the definition of the z-transform, show that $X(z) = X\left(\frac{1}{z}\right)$. **[3]**

Answer

$g[n] = x[-n]$

$G(z) = \sum_{n=-\infty}^{\infty} g[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[-n]z^{-n} = \sum_{n=-\infty}^{\infty} x[n]\,z^{n} = \sum_{n=-\infty}^{\infty} x[n]\left(\frac{1}{z}\right)^{-n}$

$= X\left(\frac{1}{z}\right)$

Therefore, it is obvious that if $x[n] = x[-n]$ then $X(z) = X\left(\frac{1}{z}\right)$.

A substantial number of students provided a correct answer to this question.

(ii) From your result in part (c) (i), show that if a pole (zero) of $X(z)$ occurs at $z = z_0$, then a pole (zero) must also occur at $z = 1/z_0$. **[3]**

Answer

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Assume that $x[n] = x[-n]$ which implies $X(z) = X\left(\frac{1}{z}\right)$.
Assume that
$X(z) = P(z)(z - z_0)$
$X\left(\frac{1}{z}\right) = P\left(\frac{1}{z}\right)\left(\frac{1}{z} - z_0\right) = P\left(\frac{1}{z}\right)(1 - zz_0)\frac{1}{z}$
For $z = \frac{1}{z_0}$, $X\left(\frac{1}{z}\right) = 0 \Rightarrow \frac{1}{z_0}$ is a zero for $X\left(\frac{1}{z}\right)$.
Same analysis is valid for poles.

A substantial number of students provided a correct answer to this question.

Verify the results in part (c) (ii) for each of the following sequences:

(iii) $\delta[n+1] + \delta[n-1]$ **[3]**

Answer
$Z\{\delta[n+1]\} = z$
$Z\{\delta[n-1]\} = z^{-1}$
$X(z) = Z\{\delta[n+1]\} + Z\{\delta[n-1]\} = z + z^{-1} = \frac{z^2+1}{z} = \frac{z^2 - i^2}{z} = \frac{(z+i)(z-i)}{z}$
$X\left(\frac{1}{z}\right) = \frac{(\frac{1}{z} + i)(\frac{1}{z} - i)}{\frac{1}{z}} = \frac{(1+zi)(1-zi)}{z} = \frac{(i-z)(i+z)}{iiz} = \frac{(z+i)(z-i)}{z}$
The zeros are $+i, -i$. We know that $\frac{1}{i} = -i$ and $\frac{1}{-i} = i$. Therefore, $\frac{1}{i}$ and $\frac{1}{-i}$ are also zeros.
The pole is 0. We see that $z = \infty$ also makes the z-transform ∞.

Only a small number of students provided all the poles of this function and the relations among them.

(iv) $\delta[n+1] - \frac{5}{2}\delta[n] + \delta[n-1]$ with $\delta[n]$ the discrete impulse function. **[3]**

$x[n] = \delta[n+1] - \frac{5}{2}\delta[n] + \delta[n-1]$
$X(z) = z + z^{-1} - \frac{5}{2} = \frac{2z^2 + 2 - 5z}{2z} = \frac{(2z-1)(z-2)}{2z}$
$X\left(\frac{1}{z}\right) = \frac{1}{z} + (\frac{1}{z})^{-1} - \frac{5}{2} = \frac{1}{z} + z - \frac{5}{2} = \frac{2z^2 + 2 - 5z}{2z} = \frac{(2z-1)(z-2)}{2z}$
Zeros are at 2 and ½ therefore their inverses are ½ and 2.

Only a small number of students provided all the poles of this function and the relations among them.

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